

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1.5+ years of experience building and deploying ML systems with LLMs, Agentic AI, and computer vision. As a Research Associate, AI at Keywords Studios, I train AI agents and build the data pipelines behind them, and was named "Rookie of the Year 2025" for beating every performance KPI. I co-authored a peer-reviewed research paper and am pursuing an M.Tech in AI and ML at BITS Pilani. I am also a marathon runner and bring the same discipline and consistency to everything I build.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon | CGPA: 8.5/10

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Awarded "**Rookie of the Year 2025**" for exceeding **every performance KPI** in model training, data pipelines, and teamwork.
- Trained and optimized **Agentic AI models** that complete complex tasks on their own, such as booking flights, placing online orders, and collecting web data, improving **task completion reliability**.
- Co-developed a **web automation AI agent** that reads the screen, follows plain English instructions, and clicks, types, and navigates websites by itself. Tested it with a **global team** to improve **action accuracy** for client releases.
- Built custom **data pipelines** and delivered **85% of the training data I generated directly to production** as the first member of a global team, speeding up model readiness.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an **AI call center system** with **LangChain** and **LangSmith** that improved response accuracy by **35%**, and deployed LLM features that reduced system latency by **25%**.
- Reduced **hallucinations (false answers)** by **20%** through **A/B tests** of prompt engineering strategies with automated LangSmith evaluation, making customer conversations more natural and trustworthy.
- Collaborated with **12 engineers** on ML workflows and was selected to present at the **Best Project by Indian Interns** initiative.

Data Analytics & ML Intern, QL2 Software – Gurgaon, Haryana

June 2024 – August 2024

- Developed **predictive ML models** for competitive pricing and automated the data collection and feature pipelines that fed them.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a **multimodal AI assistant** with the **Gemini API** that analyzes the screen in real time and gives continuous voice feedback.
- Developed a **FastAPI REST API** for model serving that handles concurrent inference requests, cutting manual task time by **40%**.
- Designed the system from scratch and optimized the **OCR pipeline** to keep **95%+ text extraction accuracy** on varied layouts.

Diabetic Optiscan

January 2025 – June 2025

- Created an AI tool that scans retinal (eye) images to detect early signs of **diabetic retinopathy**, served through a Dockerized FastAPI REST API, and co-authored a **peer-reviewed research paper** on its design and results.
- Fine-tuned a **Vision Transformer (ViT-B/16)** on the APTOS dataset and designed a custom wavelet image processing technique that improved **detection precision by 50%** over older methods.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an **image classification** model on **200,000+ banknote images** with **96% accuracy** and deployed it in a mobile app with voice feedback to help prevent fraud.
- Conducted **user interviews with visually impaired people** to refine the app's accessibility features for everyday use.

SKILLS

Languages & Frameworks: Python, C++, SQL, Git, PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, LangChain, LangGraph

AI & Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Embeddings, Semantic Search, Vector Databases (FAISS/Chroma),

Fine-tuning (LoRA, QLoRA), Prompt Engineering, Deep Learning, Supervised Learning, Computer Vision, NLP, Transformers, BERT

MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, FastAPI, REST APIs, Model Serving, Model Monitoring, Model Evaluation

Data & Cloud Tools: AWS, GCP (Vertex AI), MySQL, MongoDB, Pandas, NumPy, Matplotlib, Seaborn, Data Pipelines, Exploratory Data Analysis, Statistical Analysis, Data Structures & Algorithms, OOP

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote sensing imagery for terrain classification and geospatial pattern analysis.